Lokesh Kumar

+91-9302335988 | hellokumarlokesh@gmail.com | GitHub Profile | LinkedIn Profile

Raipur, Chhattisgarh

Technical Skills and Interests

Programming Languages: Python, C/C++, JavaScript, SQL, HTML, CSS
AI/ML & Data Science: Machine Learning, Deep Learning, NLP, Generative AI, Data Analysis, Model Deployment
Frameworks & Libraries: FastAPI, Django, Flask, Streamlit, ReactJS, TensorFlow, Scikit-learn, Pandas, NumPy
Databases & Cloud: MySQL, MongoDB, Firebase, AWS Basics
Developer Tools: Git, GitHub, VS Code, Docker, Postman
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object Oriented Programming, Database Management Systems, Software Engineering
Areas of Interest: Artificial Intelligence, Machine Learning, LLMs, RAG Systems, Full-Stack Development, Data Science
Soft Skills: Problem Solving, Analytical Thinking, Team Collaboration, Adaptability, Presentation Skills

Experience

• **AI Engineer Internship** *June - August 2025*
Apana Time Online

- Developed and enhanced an AI-powered Data Analyst Bot using FastAPI, Streamlit, and Large Language Models for intelligent data analysis and conversational querying.
- Implemented features including SQL Query Generation, contextual RAG pipelines, automated data augmentation, session-based memory handling, and file upload support for CSV, PDF, and image datasets.
- Built and integrated REST APIs, backend processing pipelines, and interactive dashboard components for real-time analytics and user-friendly data exploration.
- Worked with Python, Pandas, LangChain, vector databases, and modern AI frameworks to optimize performance, improve response accuracy, and streamline analytical workflows.

Personal Projects

• **Enterprise RAG Model**
A scalable Retrieval-Augmented Generation system built for enterprise document intelligence.

- Designed a domain-specific RAG pipeline using FAISS vector store and HuggingFace embeddings to retrieve context from large-scale proprietary document corpora with high precision.
- Implemented hybrid retrieval combining dense semantic search and BM25 sparse search, improving recall by handling both conceptual and exact-match queries across enterprise knowledge bases.
- Enforced role-based access control (RBAC) at the retrieval layer using metadata filtering, ensuring secure, compliance-ready document access across multi-tenant enterprise environments.
- Technology Used: Python, FastAPI, LangChain, FAISS, HuggingFace, Groq (Llama), Streamlit, Docker.

• **Driver Drowsiness Detection**
A real-time computer vision system that detects driver drowsiness and triggers alerts to prevent accidents.

- Implemented real-time eye and facial landmark detection using OpenCV and Dlib to continuously monitor Eye Aspect Ratio (EAR) and trigger audio alerts upon drowsiness detection.
- Technology Used: Python, OpenCV, Dlib, NumPy, Pygame.

• **Potato Disease Classification Model**
Deep learning-based web application for detecting and classifying potato leaf diseases using image classification techniques.

- Built a CNN-based potato disease classification system to identify diseases such as Early Blight and Late Blight from leaf images.
- Developed and deployed the application using FastAPI, Streamlit, Docker, TensorFlow, and Python for real-time disease prediction.
- Implemented image preprocessing, model inference, and interactive UI for farmers and agricultural use cases.
- Technology Used : Python, TensorFlow, FastAPI, Streamlit, Docker, CNN, HTML, CSS.

Education

• **Bachelor of Technology in Artificial Intelligence and Data Science** *2022-26*
Chouksey Engineering College, Bilaspur CGPA/CPI: 8.55

Achievements

• **4th Place Winner - AlgoVision Data Analytics Hackathon** KOED x IIT Roorkee *2025*

- Competed in a national-level data analytics and machine learning competition with participants from multiple institutes across India.
- Developed data-driven analytical solutions and secured 4th position among competing teams through effective problem-solving and model development.